

Kushagra Shrivastava

[Email](#) [LinkedIn](#) [Website](#)

Skills and Interests	Observational Astronomy, Cosmology, Computational Modeling, Data Analysis, Science Communication, Competition Organisation	
Education	National University of Singapore <i>Bachelor of Science in Physics (Hons.)</i> <i>Specialisation in Astrophysics</i> <i>Special Program in Science</i> <i>NUS College</i> Year 3 Sem 2 GPA: 4.77/5.00	2023 — Present
	University of Copenhagen (Københavns Universitet) <i>Semester-long Exchange Student</i> Same academic information as above	2026
Research Experience	Optimisation of LEO satellites Doppler shift compensation by reducing TLE errors and improving trajectory prediction algorithms <i>Satellite Technology and Research Center, Singapore</i> Advisor: A. Prof Abhishek Rai	2025
	Fano Resonant Metasurfaces for Virus Testing Application <i>A*STAR Institute of High-Performance Computing, Singapore</i> Advisor: Dr. Hong-Son Chu	2021—2022
	Growth Analysis of Sunspot in AR12741 using High-Resolution Solar Imaging <i>National Junior College, Singapore</i> Advisors: Mr. Alfred Tan and Ms. Ning Hwee Tiang	2021
	Closing-up onto our sun: Solar Imaging <i>National Junior College, Singapore</i> Advisors: Mr. Alfred Tan and Ms. Ning Hwee Tiang DOI: https://doi.org/10.1051/epjconf/202024007011	2019—2020
Research Schools	GraSPA (Graduate School for Particle and Astroparticle Physics) <i>Laboratoire d'Annecy de Physique des Particules, Annecy, France</i>	2026
Teaching and Mentoring	CS1010S Introduction to Programming Methodology <i>Teaching Assistant</i> National University of Singapore	2025 — Present
	SP2273 Working on Interdisciplinary Science, Pythonically <i>Mentor and Teaching Assistant</i> National University of Singapore	2025 — Present
	SP3176 The Universe <i>Mentor and Teaching Assistant</i> National University of Singapore	2025 — Present
	SP3275 Science for a Sustainable Earth <i>Mentor and Teaching Assistant</i> National University of Singapore	2025 — Present
Professional Memberships	Royal Astronomical Society <i>Elected Fellow (FRAS)</i>	2023—Present
	European Astronomical Society <i>Affiliated Member</i>	2023—Present

Leadership Positions	Cosmic Cup	2023—Present
	<i>Founder, Administrator, Question Master</i>	
	Long-Type Questions: Explosion and Expansion — Hubble Parameter calculation through Type Ia Supernovae, Vedic Knowledge — Practical Astronomy through Astrology's lens	
	Website: www.cosmiccup.org	
	National University of Singapore Astronomical Society	
	<i>President, 27th Executive Committee</i>	2024—2025
	<i>Director of Academia, 26th Executive Committee</i>	2023—2024
	AstroChallenge	
	<i>Finals In-Charge</i>	2025
	<i>Question Master</i>	2024—2025
Scholarships and Awards	Long-Type Questions: MOND-ifynig Newton's Gravitation (2024), The Sound of Space — Baryonic Acoustic Oscillations (2025)	
	Website: www.astrochallenge.org	
	CodeDay	
	<i>Assistant Regional Manager, Singapore</i>	2023
	<i>Volunteer, Lucknow, India</i>	2024—2025
	<i>Volunteer, Kolkata, India</i>	2024
	Website: www.codeday.org	
	NUS Science and Technology Merit Scholarship	2023—2027
	<i>Awarded by National University of Singapore</i>	
	Full-ride scholarship (with other benefits) for undergraduate studies.	
	A*STAR Youth Scholarship	2019—2022
	<i>Awarded by A*STAR and Ministry of Education, Singapore</i>	
	Full-ride scholarship (with other benefits) for Grade 9 to Grade 12 studies.	
	Best Research Presentation at SPS Congress	2025
	<i>Awarded by Special Program in Science</i>	
	Awarded for best presentation on “Optimisation of LEO satellites Doppler shift compensation by reducing TLE errors and improving trajectory prediction algorithms”	
	Finalist, International Natural Sciences Tournament	2024
	<i>Awarded by International Natural Science Tournament Committee</i>	
	Second Runner-up, PLANCKS Singapore Preliminaries	2024
	<i>Awarded by Institute of Physics, Singapore, Student Chapter</i>	
	Sembcorp Marine Prize	2023
	<i>Awarded by National University of Singapore</i>	
	Awarded for being the “Best Student in Physics”	
	Best Research in Sustainable Material Category	2022
	<i>Awarded by National Junior College</i>	
	Awarded for “Fano Resonant Metasurfaces for Virus Testing Application” research.	
	Gold Prize, International Geosciences Challenge	2022
	<i>Awarded by Asia Oceania Geosciences Society</i>	
	Awarded for “Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging” research presentation.	
	Merit and Commendation Awards, NJC STaR Symposium	2021
	<i>Awarded by National Junior College</i>	
	Awarded for “Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging” research.	

	Merit Award, NJC STaR Symposium	2020
	<i>Awarded by National Junior College</i>	
	Awarded for “Closing-up onto our sun: Solar Imaging” research.	
Conferences	NUS Special Program in Science Research Congress	2025
	Presented <i>Optimisation of LEO satellites Doppler shift compensation by reducing TLE errors and improving trajectory prediction algorithms.</i>	
	NJC STaR Symposium (Virtual)	2022
	Presented <i>Fano Resonant Metasurfaces for Virus Testing Application.</i>	
	Singapore Science and Engineering Fair (Virtual)	2022
	Presented <i>Fano Resonant Metasurfaces for Virus Testing Application.</i>	
	Thai Astronomical Conference (Virtual)	2022
	Presented <i>Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging.</i>	
	Singapore Science and Engineering Fair (Virtual)	2021
	Presented <i>Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging.</i>	
	Asia-Oceania Geosciences Society AGM (Virtual)	2021
	Presented <i>Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging.</i>	
	NJC STaR Symposium (Virtual)	2021
	Presented <i>Growth Analysis of Sunspot in AR12741 Using High-Resolution Solar Imaging.</i>	
	NJC STaR Symposium	2020
	Presented <i>Closing-up onto our sun: Solar Imaging.</i>	
	South-East Asian Astronomy Networking Meeting	2019
	Presented <i>Closing-up onto our sun: Solar Imaging.</i>	